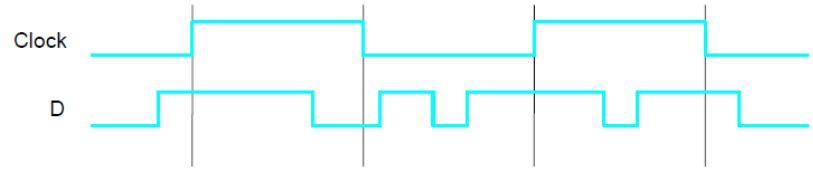
Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

Problem Set 6

1. Consider the following timing diagram. Assuming that the *D* and *Clock* inputs shown are applied to the circuit in Figure 7.12 of the textbook, draw waveforms for the Q_a, Q_b, and Q_c signals.

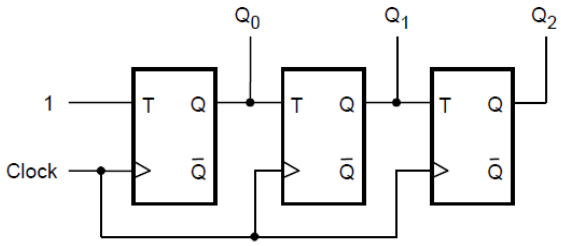


2. Show how a JK flip-flop can be constructed using a T flip-flop and other logic gates.

3. Write a behavioural VHDL code that represents a T flip-flop with an asynchronous clear input.

4. Give the expressions for the D inputs of a 4-bit universal shift register, built from D flip-flops, which can shift in both the left-to-right and right-to-left directions, and it has parallel-load capability. Assume in a left-to-right shift operation, a zero is entered into the shift register from the left hand side. Similarly, in a right-to-left shift operation, a zero is entered into the shift register from the right hand side.

5. What is the sequence that the following circuit counts in?



6. Given a 100-MHz clock signal, derive a circuit using D flip-flops to generate 50-MHz and 25-MHz clock signals. Draw a timing diagram for all three clock signals, assuming reasonable delays.

7. Design a three-bit synchronous up/down counter using T flip-flops. It should include a control input called $\overline{Up}/Down$. If $\overline{Up}/Down = 0$, then the circuit should behave as an up-counter. If $\overline{Up}/Down = 1$, then the circuit should behave as a down-counter.

8. Repeat Q7 using D flip-flops.

9. Write behavioural VHDL code that represents a 24-bit up/down counter with parallel load and asynchronous reset.

10. Write the structural VHDL code for the circuit in Figure 7.28 of the textbook. Use the counter in Figure 7.25 as a sub-circuit.

11. Determine the maximum clock frequency that can be used for the circuit in Figure 7.25 of textbook. Use the timing parameters given in Section 7.15.